

Vadodara Institute of Engineering

MIA ATL Report for Phase -1 (9-1-2026 to 8-4-2026)

Name of School	Fertilizer Nagar
Principal of School	Bharat Jadav
Nodal Teacher of School	Shaunak Thakar
RMOC	Dr.Sudhir Joshi
VIER Mentor Faculty	Naman Shah
Core Team Leader	Jayraj Chavda
VIER Interns	(1) Sujal Rana (2) Ayan Parmar (3) Dhyey Shah
Total ATL visit days	(1) 10 (2) 5 (3) 1
Total Atal sessions	(1) 10 (2) 5 (3) 1

(A) Projects

Titles & Objectives	- IoT connects smart devices across sectors like healthcare, agriculture, and smart cities, enabling real time monitoring, automation, and data driven decisions that improve efficiency, safety, and resource management for individuals and organizations. - Students should learn IoT applications, security challenges, and innovation opportunities to build future ready skills in modern world.
Use of Hardware / Components / Tools	- Arduino Uno and NodeMCU ESP8266 are popular platforms used in IoT projects to control devices, process data, and enable wireless communication between systems. - Sensors like temperature, humidity, and motion sensors collect real world data, which is processed by these boards to automate tasks and create smart applications in various fields.
Use of Software	- Arduino IDE is used to write and upload code to microcontrollers, helping students develop IoT projects and understand how hardware and software interact in real-world applications. - Code.org provides beginner-friendly coding lessons and activities, allowing students to learn programming concepts and build a strong foundation for creating IoT-based solutions.
Suggestions / Improvement / Solution provided	Make real life projects Smart Irrigation System Automatically waters plants based on soil moisture levels, saving water and improving crop growth.

	• Home Automation System Control lights, fans, and appliances remotely using a mobile app via Wi-Fi. • Smart Dustbin Uses ultrasonic sensors to open the lid automatically and notify when the bin is full. • Weather Monitoring System Collects temperature, humidity, and pressure data and displays it online in real time.
(B) Innovative topics / technologies discussed: Machine Learning , Android, Artificial Intelligence.	

Name & Sign of VIER Mentor Faculty with date: Naman Shah 20/04/2026